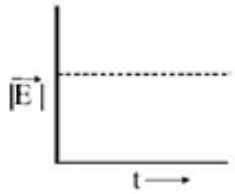
Quizrr Chapter-Wise Test for JEE Main - 2025

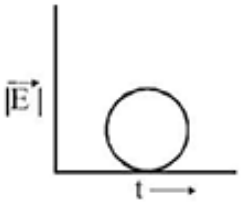
Q.1

Which of the following diagrams represent the variation of electric field vector with time for a circularly polarised light?

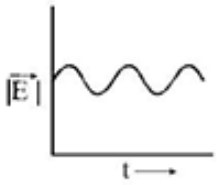
(A)



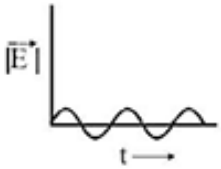
(B)



(C)

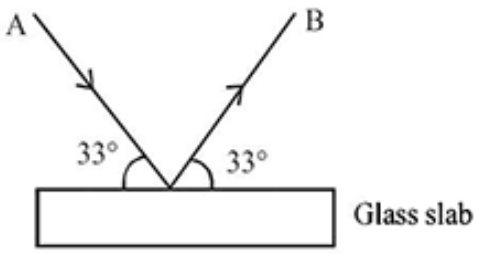


(D)



Q.2

A beam of light is incident on a glass slab $(\mu = 1.54)$ in a direction as shown in the figure. The reflected light is analysed by a polaroid prism. On rotating the polaroid, (tan $57^\circ = 1.54$)



(A) the intensity remains unchanged

(B) the intensity is reduced to zero and remains at zero

(C) the intensity gradually reduces to zero and then again increase

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(D) the intensity increases continuously

Q.3

Two polaroids are placed in the path of unpolarized beam of intensity I_0 such that no light is emitted from the second polaroid. If a third polarioid whose polarization axis makes an angle θ with the polarization axis of first polaroids, is placed between these polaroids then the intensity of light emerging from the last polaroid will be

(A) $\left(\frac{I_0}{8}\right) \sin^2 2\theta$

(B) $\left(\frac{I_0}{4}\right) \sin^2 2\theta$

(C) $\left(\frac{I_0}{2}\right) \cos^4 \theta$

(D) $I_0 \cos^4 \theta$

Q.4

Abeam of unpolarised light of intensity I_0 is passed through a polaroidAand then through another polaroid B which is oriented so that its principal plane makes an angle of 45° relative to that of A. The intensity of the emergent light is

(A) I_0

(B) $I_0/2$

(C) $I_0/4$

(D) $I_0/8$

Q.5

Two beams, A and B, of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity), a rotation of polaroid through 30° makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and
I_B respectively, then $\frac{I_A}{I_B}$ equals

(A) 3

(B) $\frac{3}{2}$

(C) 1

(D) $\frac{1}{3}$

Q.6

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Two polaroids are placed in the path of unpolarized beam of intensity I_0 such that no light is emitted from the second polaroid. If a third polarioid whose polarization axis makes an angle θ with the polarization axis of first polaroid, is placed between these polaroids then the intensity of light emerging from the last polaroid will be

(A) $\left(\frac{\mathrm{I}_0}{8}\right) \sin^2 2\theta$

(B) $\left(\frac{\mathrm{I}_0}{4}\right) \sin^2 2\theta$

(C) $\left(\frac{\mathrm{I}_0}{2}\right) \cos^4 \theta$

(D) $\mathrm{I}_0 \cos^4 \theta$

Q.7

An unpolarised beam of intensity I_0 is incident on a pair of nicols making an angle of 60° with each other. The intensity of light emerging from the pair is

(A) I_0

(B) $I_0/2$

(C) $I_0/4$

(D) $I_0/8$

Q.8

The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refractive index n), is

(A) $\tan^{-1}(1/n)$

(B) $\sin^{-1}(1/n)$

(C) $\sin^{-1}(n)$

(D) $\tan^{-1}(n)$

Q.9

A polaroid is placed at 45° to an incoming light of intensity I. Now the intensity of light after polarisation would be

(A) I

(B) I/2

(C) $\mathrm{I}/\sqrt{2}$

(D) zero

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Q.10

Unpolarised light is incident on a dielectric of refractive index $\sqrt{3}$. What is the angle of incidence if the reflected beam is completely polarised?

(A) 30°

(B) 45°

(C) 60°

(D) 75°

Q.11

When an unpolarized light of intensity I_0 is incident on a polarizing sheet, the intensity of the light which does not get transmitted is

(A) $\frac{1}{4}I_0$

(B) $\frac{1}{2}I_0$

(C) I_0

(D) zero

Q.12

The angle of polarisation for any medium is 60°. The critical angle for this is

(A) $\sin^{-1}\frac{1}{\sqrt{3}}$

(B) $\cos^{-1}\sqrt{3}$

(C) $\sin^{-1}\sqrt{3}$

(D) $\tan^{-1}\sqrt{3}$

Q.13

If I_0 is the intensity of the principal maximum in the single slit diffraction pattern, then what will be its intensity when the slit width is doubled?

(A) $4I_0$

(B) $2I_0$

(C) $\frac{I_0}{2}$

(D) I_0

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Q.14

Two polaroids have their polarizing directions parallel so that the intensity of a transmitted light is maximum. The angle through which either polaroid must be turned if the intensity is to drop by one-half is

(A) 135°

(B) 90°

(C) 120°

(D) 180°

Q.15

A mixture of polarised and unpolarised light is incident on a polariser. As the polariser is rotated through 360°, it is found that the minimum and the maximum of the transmitted intensity is in the ratio $1 : q$. The ratio of intensities of polarised to unpolarised light in the incident beam is

(A) $\frac{1}{2}(q-1) : 1$

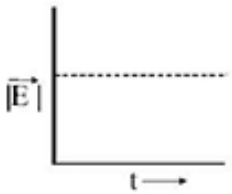
(B) $1 : \frac{1}{2}(q-1)$

(C) $\frac{1}{2}q : 1$

(D) $1 : \frac{1}{2}q$

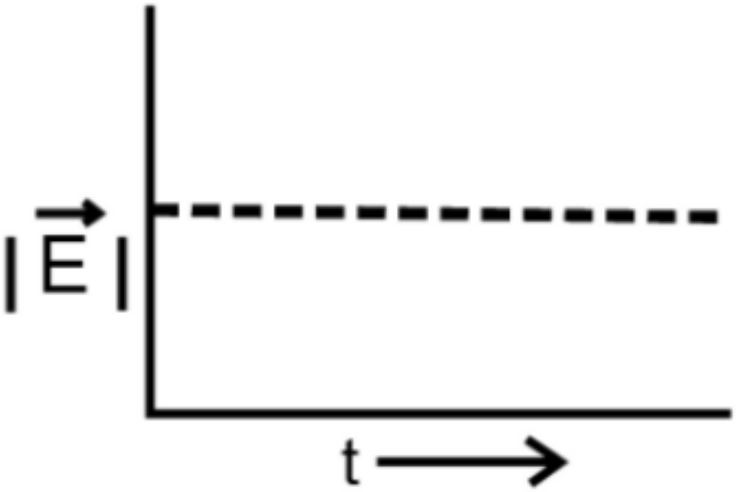
Answers & Solutions

Q.1 Answer:



Solution:

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When two plane - polarised waves are superimposed, then under certain conditions the resultant light vector rotates with a constant magnitude in a plane perpendicular. The tip of the vector traces a circle and the light is said to be circularly polarised. To form circularly polarised light

$$\mathrm{E_x} = \mathrm{E_0} \sin \omega \mathrm{t}$$
$$\mathrm{E_y} = \mathrm{E_0} \cos \omega \mathrm{t} = \mathrm{E_0} \sin \left(\omega \mathrm{t} + \tfrac{\pi}{2}\right)$$

Where $\mathrm{E_0}$ is amplitude

Resultant amplitude

$(\vec{E})^2 = E_0 + E_0 + 2E_0E_0 \cos \frac{\pi}{2}$

$|\overrightarrow{\mathrm{E}}|^2 = \mathrm{E_0}\sqrt{2} =$ **constant**

Q.2 Answer:

the intensity gradually reduces to zero and then again increase

Solution:

Here Angle of incidence, $\mathrm{i} = 57 \tan 57^\circ = 1.54$

$\mathrm{u_{glass}} = \tan$

It means, Here Brewster's law is followed and the reflected ray is completely polarised. Now, when reflected ray is analysed through a polaroid then intensity of light is given by malus law.

i.e. $I = I_0 \cos^2 \theta$

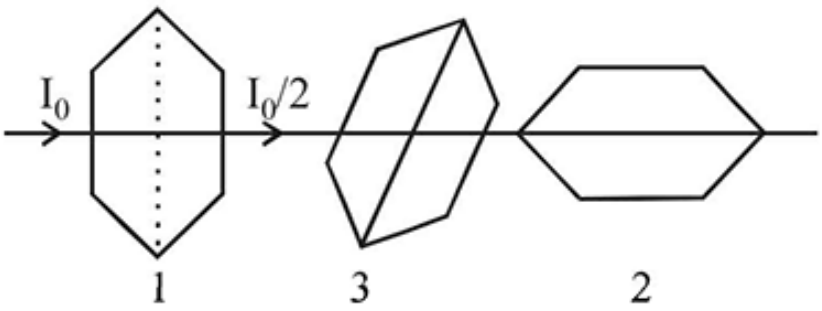
on rotating polaroid ' θ **' changes. Due to which intensity first decreases and then increases.**

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Q.3 Answer:

$\left(\frac{I_0}{8}\right) \sin^2 2\theta$

Solution:



The intensity of light transmitted through third polaroid,

$I = \frac{I_0}{2} \cos^2 \theta$

$\therefore$ intensity of light transmitted through the last polaroid

$I' = \left(\frac{I_0}{2} \cos^2 \theta\right) \cdot \cos^2(90 - \theta) = \frac{I_0}{2} \cos^2 \theta \cdot \sin^2 \theta$

$= \frac{I_0 \times 4 \sin^2 \theta \cdot \cos^2 \theta}{2 \times 4} = \frac{I_0}{8} \times (2 \sin \theta \cdot \cos \theta)^2$

$= \left(\frac{I_0}{8}\right) \sin^2 2\theta$

Q.4 Answer:

$I_0/4$

Solution:

$I \alpha A^2$ where A is the amplitude and I is the intensity

For unpolarised light, it can be resolved into two components of equal magnitude in any pair of perpendicular directions. So the amplitude that is along the plane of polarisation of the polaroid will always be $\frac{A}{\sqrt{2}}$ So, the intensity which will pass through it will always be $\frac{I_0}{2}$ Now when this polarised light falls on polaroid B whose plane of polarisation makes an angle θ with the plane of polarisation of incident light, the component of amplitude along the plane of polarisation of the polaroid is $A \cos \theta$.

Therefore, intensity of light that passes through is $I_i \cos^2 \theta$ where I_i is the intensity of incident light.

Here, $I_i = \frac{I_0}{2}$ and $\theta = 45^\circ$

So intensity of light emerging through B is $\frac{I_0}{2} \times \frac{1}{2} = \frac{I_0}{4}$

Q.5 Answer:

$\frac{1}{3}$

Solution:

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According to malus law, intensity of emerging beam is given by, $I = I_0 \cos^2 \theta$

Now, $I_{A'} = I_A \cos^2 30^\circ$

$I_{B'} = I_B \cos^2 60^\circ$ **As** $I_{A'} = I_{B'}$

$\Rightarrow \quad I_A \times \frac{3}{4} = I_B \times \frac{1}{4} = \frac{I_A}{I_B} = \frac{1}{3}$

Q.6 Answer:

$\left(\frac{\mathrm{I}_0}{8}\right) \sin^2 2\theta$

Solution:

No light is emitted from the second polariod, so P_1 **and** P_2 **are perpendicular to each other**

Let the initial intensity of light is I_0**. So Intensity of light after transmission from first polaroid** $= \frac{\mathrm{I}_0}{2}$

Intensity of light emitted from $\mathrm{P}_3 \mathrm{I}_1 = \frac{\mathrm{I}_0}{2} \cos^2 \theta$

Intensity of light transmitted from last polaroid i.e. from

$$\mathrm{P}_2 = \mathrm{I}_1 \cos^2 (90^\circ - \theta) = \frac{\mathrm{I}_0}{2} \cos^2 \theta \cdot \sin^2 \theta$$
$$\mathrm{P}_2 = \frac{\mathrm{I}_0}{8} (2 \sin \theta \cos \theta)^2 = \frac{\mathrm{I}_0}{8} \sin^2 2\theta$$

Q.7 Answer:

$I_0/8$

Solution:

After first prism, intensity $= I_0/2$

After second prism, intensity $= (I_0/2) \cos^2(\theta)$

$= I_0/8$

Q.8 Answer:

$\tan^{-1}(n)$

Solution:

(d) According to Brewster's law of polarization, $n =$ **tanip where** i_p **is angle of incidence**

Therefore, $i_p = \tan^{-1}(n)$.

Q.9 Answer:

I/2

Solution:

$$\mathrm{I}' = \mathrm{I} \cos^2 \theta = \mathrm{I} \cos^2 (45^\circ) = \mathrm{I} \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{\mathrm{I}}{2}.$$

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Q.10 Answer:

60°

Solution:

$\mu = \tan i$
$\Rightarrow \mathrm{i} = \tan^{-1}(\mu) = \tan^{-1}(\sqrt{3}) = 60^\circ$

Q.11 Answer:

$\frac{1}{2} I_0$

Solution:

Intensity of polarized light $= \mathrm{I}_0/2$

Therefore, Intensity of light not transmitted

$= \mathrm{I}_0 - \mathrm{I}_0/2 = \mathrm{I}_0/2.$

Q.12 Answer:

$\sin^{-1} \frac{1}{\sqrt{3}}$

Solution:

$\mu = \tan \mathrm{i}_\mathrm{p} = \tan 60^\circ = \sqrt{3}$
$\sin \mathrm{i}_\mathrm{C} = \frac{1}{\mu} = \frac{1}{\sqrt{3}} \Rightarrow \mathrm{i}_\mathrm{C} = \sin^{-1} \frac{1}{\sqrt{3}}$

Q.13 Answer:

$4 I_0$

Solution:

$I = I_0 \left(\frac{\sin \phi}{\phi} \right)$ **and** $\phi = \frac{\pi}{\lambda}(b \sin \theta)$

When the slit width is doubled, the amplitude of the wave at the centre of the screen is doubled, so the intensity at the centre is increased by a factor 4**.**

Q.14 Answer:

135°

Solution:

For $I = \frac{I_0}{2}$ **and** $I = I_0 \cos^2 \theta = \frac{I_0}{2}$
$\therefore \theta = 45^\circ$
Therefore the angle through which either polaroids turned is $135^\circ \ (= 180^\circ - 45^\circ)$

Q.15 Answer:

$\frac{1}{2}(\mathrm{q} - 1) : 1$

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Solution:

Malus law is mathematically stated as, $\mathrm{I_1 = I_0\,Cos^2\,q}$

where θ **is the angle between the polarization direction of the light and the transmission axis of the polarizer.**

$\mathrm{I_0}$ **is incident intensity on polarized light**

I am the transmitted intensity

$\mathrm{I = I_0\,Cos^2\,360}$ **i.e the value of this equation is** 1**.**

$\mathrm{I = I_0}$

Hence the value of unpolarized light is 1

For polarised light;

According to malus law,

When unpolarized light is polarized with two polarizers, the intensity becomes $\mathrm{I = I_0 cos\,2(\theta) I = I_0 cos\,2(\theta)}$ **(Malus's law). But when unpolarized light is polarized with only one polarizer, the intensity is reduced to half the intensity of the unpolarized light.**

$\mathrm{I = \frac{1}{2\pi}\int_0^{2\pi} I_0\,Cos(\theta)^2 = \frac{1}{2}}$**. Hence the value of polarised is** $\frac{1}{2}$**.**

Since the intensity is reduced by half the value of q **i.e polarising angle will be subtracted by 1 . Hence its value will become** $\frac{1}{2}(\mathrm{q} - 1)$**.**

Thus the ratio of intensities of polarised to unpolarised light in the incident beam is $\frac{1}{2}(\mathrm{q} - 1) : 1$

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